

A Market World Model for Public LLMs: Complete, Honest, Auditable Crypto Cognition before Decision

Anonymous Author(s)

Abstract

Most quant stacks start from *strategy*. Public large language models (GPT-, DeepSeek-, GLM-, and Gemini-class agents) that act in cryptocurrency markets instead fail earlier: they lack a complete, honest, auditable *market cognition base*. Raw feeds are asynchronous, intermittently missing, and quality-heterogeneous; ungated tool dumps invite overconfident action. We build a *financial world-model runtime* that compiles multi-band evidence into a point-in-time (PIT) world bundle for public LLMs. The primitive is an epistemic observation $O_{j,t} = (x, \tau, q, g, r)$; a compilation operator Π_t yields $\mathcal{F}_t^{\text{AI}}$; completeness, honesty (B, U, H) , and WMI/ACWMI gate abstention; evidence IDs make actions auditable. On a real crypto PIT panel, four live public LLMs via an OpenAI-compatible gateway (temperature 0, frozen prompts) receive Compiled vs. Raw bundles on 100 OOS asset-days each. Under the typed Compiled contract, every model abstains on thin worlds (thin-world abstain 1.0); under Raw, abstain rates fall to 0.04–0.75 and models take directional bets into sparse support. An Ungated ablation (same content fields, numeric WMI, *no* hard abstain flag; 50 days) yields mean thin-world abstain 0.74 (range 0.38–0.94)—disclosure helps, but the typed boolean is what enforces full refusal. Mean ΔCE (Compiled–Raw) is secondary—not a strategy Holy Grail.

CCS Concepts

• **Computing methodologies** → **Artificial intelligence**; *Machine learning*; • **Applied computing** → *Economics*.

Keywords

world models, public LLMs, AI agents, cryptocurrency, point-in-time, abstention, trustworthy AI, market cognition

ACM Reference Format:

Anonymous Author(s). 2026. A Market World Model for Public LLMs: Complete, Honest, Auditable Crypto Cognition before Decision. In *Proceedings of the 7th ACM International Conference on AI in Finance (ICAIF '26)*. ACM, New York, NY, USA, 4 pages. <https://doi.org/10.1145/XXXXXXX.XXXXXX>

1 Introduction

Public LLMs are increasingly asked to analyze or trade crypto markets. The bottleneck is not another alpha signal. It is *understanding*: before decision, an agent needs a market world that is **complete** (missingness disclosed), **honest** (stale or role-incoherent evidence gated), and **auditable** (actions bound to evidence). Venue fragmentation, derivatives, on-chain flows, news, and macro vintages arrive on incompatible clocks [12, 14]. Generative world models [6, 7, 10] and tool-using agents [1, 19] typically assume a usable observation stream. Crypto agents often do not have one.

Thesis. We build a *market world-model runtime* for public LLMs: compile raw multi-band evidence into a PIT-safe cognition bundle that GPT / DeepSeek / GLM / Gemini-class models can call. Strategy metrics are secondary probes that the world has content—not the product objective. In slogan form: *quant from strategy; this runtime from understanding*.

What we claim (and do not). RQ1 tests whether a typed world interface can *enforce* world-conditional abstention across live public LLMs—a runtime contract, not a claim that models spontaneously discover thinness from raw ticks. RQ2 tests whether compiled band fields are economically nonempty versus same-input momentum. We do not claim unconditional trading alpha or a Dreamer-style generative simulator.

Contributions.

- (1) **Cognition semantics.** Epistemic observations, compilation Π_t , a lag reconstruction bound, completeness/honesty fields, WMI/ACWMI abstention, evidence-bound actions.
- (2) **Runtime for public LLMs.** Vintage-aware collectors, Band-PIT previous-close clock, quality-tagged bundles, availability shocks O_t , frozen Compiled / Ungated / Raw protocol with an OpenAI-compatible multi-vendor adapter.
- (3) **Live understanding-first validation.** Four live public LLMs abstain under the Compiled contract when the world is thin and often trade under Raw; an Ungated ablation removes the hard abstain flag; a secondary content probe shows vintaged band fields are economically nonempty.

2 Related Work

World models in AI. World models learn compact states and often dynamics for control or imagination [6, 7, 10]. We address the complementary *observation-side* problem: a PIT-safe, quality-tagged crypto world *before* dynamics or LLM policy. We do not claim a Dreamer/JEPA-style generative simulator; we claim the missing systems layer those models usually assume.

LLM agents in finance. Recent work builds LLM trading agents and finance-tuned models—return prediction with ChatGPT [13], open financial LLMs [18], memory-augmented trading agents [11, 20], and tool-augmented multimodal trading agents [21]. Agentic workflows call tools over streams [19]. These systems typically assume the observation stream is already aligned and trustworthy. We evaluate within-model Compiled vs. Raw (and Ungated) under a frozen schema so the estimand is the *world interface*, not a new chat prompt or memory module.

Selective prediction and trustworthy AI. Abstention can be optimal under noise [3, 4]. We tie refusal to world quality (WMI/ACWMI) via an explicit runtime contract, not classifier confidence alone.

PIT measurement and crypto microstructure. Macro vintages [2] and crypto microstructure [12, 14] motivate multi-band compilation.

ICAIF '26, Milan, Italy

2026. ACM ISBN 978-1-4503-XXXX-X/26/11
<https://doi.org/10.1145/XXXXXXX.XXXXXX>

Bootstrap discipline for secondary economic probes follows [8, 16, 17]. Asset-pricing ML [5, 9, 15] is adjacent but not our estimand: we validate that the compiled world is nonempty for agents, not that a return factor is priced.

3 Market World-Model Semantics

DEFINITION 1 (EPISTEMIC OBSERVATION). *An epistemic observation is $O_{j,t} = (x_{j,t}, \tau_{j,t}, q_{j,t}, g_{j,t}, r_{j,t})$: value, latest-available time, quality, main-view gate, and semantic role. A bare feature dump that omits (τ, q, g, r) is not a world observation.*

DEFINITION 2 (FINANCIAL WORLD STATE). *A financial world state at t is a quality-tagged, PIT-safe compilation $W_t = \Pi_t(\mathcal{F}_t^{\text{raw}})$ together with a scalar world quality $q(W_t)$ (e.g. WMI/ACWMI) that gates abstention; $\mathcal{F}_t^{\text{AI}} = \sigma(W_t)$.*

DEFINITION 3 (COMPILATION). $\Pi_t = B_t \circ M_t \circ A_t$: align clocks, apply honesty/missingness gates, assemble band views.

Completeness, honesty, quality. Ready/limited/missing counts disclose coverage. Breadth/stability/honesty give $\text{WMI}_t = B_t U_t H_t$; ACWMI supports regime-conditional gating. The runtime exposes `should_ai_abstain` and `thin_world` as first-class fields in the Compiled contract.

ASSUMPTION 1 (BOUNDED LAG AND INCREMENTS). *Each band j has a maximum observation lag Δ_j under the decision clock, and the latent state S_t has bounded increments $\|S_u - S_{u-1}\| \leq \bar{\delta}$.*

PROPOSITION 1 (LAG RECONSTRUCTION BOUND). *Under Assumption 1, a reconstruction $\tilde{S}_t$ built from lagged, noisy, and partially missing observations satisfies*

$$\|\tilde{S}_t - S_t\| \leq \underbrace{C_{\text{del}}\bar{\delta}}_{\text{delay}} + \underbrace{C_{\text{noi}}\epsilon_t}_{\text{noise}} + \underbrace{C_{\text{miss}}m_t}_{\text{missingness}},$$

where ϵ_t aggregates observation noise and m_t aggregates disclosed missing mass. Compilation cannot erase delay; it can only refuse to pretend the bound is zero.

PROPOSITION 2 (COMPILATION $\neq$ FEATURE DUMP). *Enlarging raw span without Π_t need not enlarge usable $\mathcal{F}^{\text{AI}}$: ungated stale evidence can raise raw volume while lowering honesty/stability.*

PROPOSITION 3 (WORLD-CONDITIONAL ABSTENTION). *If non-abstain actions exceed a world-dependent cost $c_{\text{abs}}(W_t)$, Bayes action is abstain; with costs decreasing in WMI, abstention is a lower set in world quality [3].*

PROPOSITION 4 (LOBO CONTENT VS. GATING). *Deleting band j changes economic value through a content channel (actions change given the same gate) and a gating channel (the abstention set changes). Empirically we report content share under an ungated rule so the probe speaks to nonempty fields, not only refusal.*

REMARK 1 (NOT A GENERATIVE WM). *We do not learn $p(s_{t+1} | s_t, a_t)$. We deliver a state compiler + abstention runtime for public LLM consumers—the observation interface generative world models usually assume.*

World-Model-First: compile before predict



latest_* · readiness · eight evidence bands · main/diagnostic separation

Figure 1: World-model runtime: raw crypto evidence → compiled cognition bundle → public LLM decision / abstain.

Table 1: World bundle served to public LLMs (cognition base).

Group	Fields
Timing	<code>decision_asof</code> , <code>previous-close</code> protocol
Completeness	<code>n_ready/limited/missing</code> , band statuses
Honesty	<code>B, U, H</code> , main-view / stale gates
Quality	WMI, ACWMI, <code>should_ai_abstain</code> , <code>thin_world</code>
Content	<code>macro_tilt</code> , <code>alt_tilt</code> , regime, cascade
Audit	<code>evidence_ids</code> , EAR required

4 Runtime for Public LLMs

Figure 1 shows the anonymized prototype. Collectors fill vintage-aware stores; BandPIT reconstructs readiness at $(t-1)$ 23:59 for payoff r_t (previous-close decision clock; no same-day look-ahead). The runtime emits a world bundle consumed by rule or LLM agents via frozen prompts (temperature 0). An OpenAI-compatible adapter routes the same schema to GPT / DeepSeek / GLM / Gemini-class endpoints; offline stylized followers keep the protocol reproducible without keys. Availability shocks O_t are queryable from collection logs as a quasi-exogenous lever on world thickness.

Three information arms.

- **Compiled (product contract):** full bundle including hard `should_ai_abstain` frozen prompt requires abstain when the flag is true.
- **Ungated (ablation):** same content, completeness, honesty, and numeric WMI/ACWMI, but *no* hard abstain boolean and no forced policy—the model must judge thinness itself.
- **Raw (without world model):** thin feed (mom5 only); no quality index, no band roles, no abstention guidance.

Understanding metrics prioritize thin-world abstain rate and evidence-audit (EAR) proxy over CE. Durable bands in the archive: {exchange, macro, alterna (ready rates $\approx 0.80/1/1$); news/on-chain/options are right-censored and disclosed as missing. Actions are restricted to {bullish, bearish, neutral, abstain, ...}

Why this is not “just instruction following.” A fair objection is that Compiled abstention is prompt obedience. We treat that as a *feature of the runtime contract*: production agents need an enforceable refuse interface when $q(W_t)$ is low (Prop. 3). The scientific contrast is then (i) Raw agents that lack any world quality signal and over-act, and (ii) Ungated agents that see the same disclosed world but must refuse without a boolean. Spontaneous refusal under Ungated is a bonus; reliable refusal under Compiled is the product requirement.

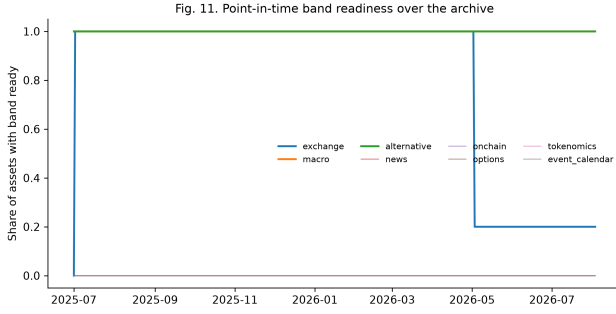


Figure 2: PIT readiness: completeness is partial—missing bands are first-class in the bundle.

Table 2: Live public LLMs on 100 OOS asset-days (temp. 0). Understanding metrics first; Δ CE secondary.

Model	Thin-abs.	Abs. C	Abs. R	Δ CE
gpt-5.4-mini	1.00	1.00	0.04	+0.73
deepseek-v4-flash	1.00	1.00	0.75	+1.22
glm-5.2	1.00	1.00	0.11	+1.04
gemini-3.5-flash-lite	1.00	1.00	0.07	+1.30
Mean	1.00	1.00	—	+1.07

5 Experiments

5.1 Setup

PIT panel ~ 399 days (≈ 13 months), 10 liquid names, previous-close clock, IS/OOS 200/200 days. Live RQ1 uses a stratified OOS sample of 100 asset-days per model for Compiled/Raw (temperature 0, frozen prompts, OpenAI-compatible multi-vendor gateway). Models: gpt-5.4-mini, deepseek-v4-flash, glm-5.2, gemini-3.5-flash-lite. Ungated ablation uses the same protocol on a stratified 50-day subsample (same seed). Offline diagnostics (stylized public-LLM follower; compiled-aware / momentum / noisy mocks) reproduce the protocol on the full OOS panel without keys. On this archive, scarce/thin labels bind on the OOS window by design—the primary test is honesty under sparse support, not a dense-world trading horse-race.

5.2 RQ1 (primary): Does the world-model contract change live LLM behavior?

Table 2 and Figure 3: under Compiled, all four live models abstain on every sampled thin-world day (thin-world abstain 1.0; EAR proxy 1.0). Rationales cite world fields (should_ai_abstain / missingness / readiness) on 93–99% of Compiled days. Under Raw—same dates/assets, no WMI guidance—abstain rates drop to 0.04–0.75 and models take directional positions (Table 3). Mean Δ CE (Compiled–Raw) is +1.07; we report it last because Compiled CE is zero by construction under full abstention—the number mainly reflects Raw overtrading losses, not a Holy Grail.

Ungated ablation (no hard flag). Table 4: same content + numeric WMI/ACWMI, soft prompt, 50 stratified OOS days, no should_ai_abstainlive models exhibit when they trade without a cognition base.

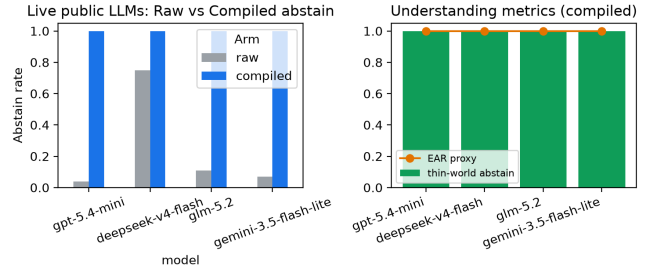


Figure 3: Live public LLMs: Compiled contract refuses thin worlds; Raw arms often trade.

Table 3: Raw-arm action mix (live, 100 days). Compiled arms are 100% abstain.

Model	Abs.	Bull.	Bear.	Neut.
gpt-5.4-mini	0.04	0.28	0.38	0.30
deepseek-v4-flash	0.75	0.14	0.04	0.07
glm-5.2	0.11	0.40	0.48	0.01
gemini-3.5-flash-lite	0.07	0.44	0.49	0.00

Table 4: Ungated ablation: thin-world abstain without hard should_ai_abstain (50 days).

Model	Abs. U	Thin-abs. U	Abs. R
gpt-5.4-mini	0.80	0.80	0.04
deepseek-v4-flash	0.82	0.82	0.75
glm-5.2	0.94	0.94	0.11
gemini-3.5-flash-lite	0.38	0.38	0.07
Mean	0.74	0.74	—

Table 5: Offline diagnostics (full OOS): world-honoring vs. strategy-like mocks.

Consumer	Thin-abs.	Δ CE	EAR
public-llm-follower	1.00	+0.202	1.00
mock-compiled-aware	1.00	+0.202	1.00
mock-momentum	0.00	0.000	1.00
mock-noisy	0.00	+0.212	1.00

Mean thin-world abstain falls from 1.0 (Compiled) to 0.74. Three models still refuse often (0.80–0.94); gemini-3.5-flash-lite drops to 0.38 and trades. Disclosure alone is helpful but incomplete; the typed runtime contract is what enforces full thin-world refusal across vendors.

Offline diagnostics. Table 5: world-honoring followers abstain at 1.0 on the full OOS panel; momentum-only and noisy mocks ignore thin-world flags (thin-world abstain 0)—the same failure mode Raw

Table 6: Secondary content probe (not the LLM product claim).

Policy / contrast	Sharpe	CE
Momentum always	0.101	−0.202
Thick ungated (content)	0.767	0.132
Mechanism – Momentum	—	0.334 ($p=0.034$)

5.3 RQ2: Is the compiled world nonempty? (content probe)

To show the bundle is not an empty label, a transparent rule shares return inputs with momentum and adds vintaged macro_tilt / alt_tilt. Pre-specified OOS contrast mechanism – momentum: $\Delta CE=0.334$ ($p=0.034$; stationary $p=0.022$). LOBO (Prop. 4): deleting macro or alternative collapses to momentum (content share 1.0 under the ungated rule). Relative gaps survive 10–25bps costs; absolute CE is cost-fragile. A 2017–2026 audit without vintaged archives finds no hidden return-rule alpha. These results *probe content*, they do not define the product.

5.4 Discussion

Quant stacks usually ask “what trades?” We ask “what does the LLM know, and should it act?” The live three-arm ladder on this sparse archive is:

$$\text{Abs}_{\text{Compiled}} = 1.0 > \text{Abs}_{\text{Ungated}} \approx 0.74 > \text{Abs}_{\text{Raw}} \in [0.04, 0.75].$$

Compiled delivers a vendor-stable refuse interface; Ungated shows that completeness/WMI disclosure alone recovers most—but not all—refusal (Gemini drops to 0.38); Raw shows what happens without a world model. Table 6 then shows compiled band fields are not decorative labels. Together: *understanding first, strategy second*.

Threats to validity. (1) *Prompt contract.* Compiled abstention is partly instructed; we disclose this and use Ungated as the soft-judgment contrast rather than hiding it. (2) *Thin archive.* Scarce rate ≈ 1 on OOS means we stress honesty, not dense-world calibration. (3) *Raw baseline strength.* Raw is intentionally thin (momentum-only), matching a common “just give the model a price signal” integration; Ungated is the stronger same-content control. (4) *Sample size.* Live Compiled/Raw uses 100 asset-days; Ungated 50; a full ~2000-row OOS sweep is supported by the harness when budget allows and would mainly tighten intervals under current scarcity.

Expanding historical band coverage is the natural next measurement step: denser worlds would let ACWMI separate “act” from “abstain” on a nontrivial fraction of days while keeping the same three-arm protocol.

6 Limitations

Live Compiled/Raw uses 100 stratified OOS asset-days; Ungated uses 50. Full-panel live replication remains future work. Sparse archives make WMI thresholds bind almost always on the sample—honest under thin support, not a dense-world calibration study. Compiled prompts include an explicit abstain rule when should_ai_abstain

is true; we report this contract openly and use Ungated as the soft-judgment contrast. News/on-chain/options history remains right-censored and is disclosed as missing. We do not claim generative dynamics $p(s_{t+1} | s_t, a_t)$ or unconditional trading alpha. Gateway routing and model IDs are vendor-dependent; prompts and schema are frozen for within-model contrasts.

7 Conclusion

Public LLMs need a market to *understand* before they decide. We compile asynchronous crypto evidence into a complete, honest, auditable world-model bundle and show that four live public LLMs abstain under the Compiled contract when the world is thin, while Raw arms often trade; Ungated ablation separates typed enforcement from soft judgment; content probes confirm the compiled state is nonempty. Strategy is downstream of understanding—not the other way around.

References

- [1] Tom Brown et al. 2020. Language Models are Few-Shot Learners. *NeurIPS* (2020).
- [2] Dean Croushore and Tom Stark. 2001. A Real-Time Data Set for Macroeconomists. *Journal of Econometrics* 105, 1 (2001), 111–130.
- [3] Ran El-Yaniv and Yair Wiener. 2010. On the Foundations of Noise-Free Selective Classification. *JMLR* 11 (2010), 1605–1641.
- [4] Yonatan Geifman and Ran El-Yaniv. 2017. Selective Classification for Deep Neural Networks. In *NeurIPS*.
- [5] Shihao Gu, Bryan Kelly, and Dacheng Xiu. 2020. Empirical Asset Pricing via Machine Learning. *Review of Financial Studies* 33, 5 (2020), 2223–2273.
- [6] David Ha and Jürgen Schmidhuber. 2018. World Models. In *NeurIPS*.
- [7] Danijar Hafner, Timothy Lillicrap, Jimmy Ba, and Mohammad Norouzi. 2020. Dream to Control: Learning Behaviors by Latent Imagination. In *ICLR*.
- [8] Campbell R. Harvey, Yan Liu, and Heqing Zhu. 2016. ... and the Cross-Section of Expected Returns. *Review of Financial Studies* 29, 1 (2016), 5–68.
- [9] Bryan T. Kelly, Seth Pruitt, and Yanan Su. 2019. Characteristics Are Covariances: A Unified Model of Risk and Return. *Journal of Financial Economics* 134, 3 (2019), 501–524.
- [10] Yann LeCun. 2022. A Path Towards Autonomous Machine Intelligence. *Open Review* (2022).
- [11] Yang Li, Yangyang Yu, Haohang Li, Zhiyang Chen, and Khaldoun Khashanah. 2023. TradingGPT: Multi-Agent System with Layered Memory and Distinct Characters for Enhanced Financial Trading Performance. *arXiv preprint arXiv:2309.03736* (2023).
- [12] Yukun Liu, Aleh Tsyvinski, and Xi Wu. 2022. Common Risk Factors in Cryptocurrency. *Journal of Finance* 77, 2 (2022), 1133–1177.
- [13] Alejandro Lopez-Lira and Yuehua Tang. 2023. Can ChatGPT Forecast Stock Price Movements? Return Predictability and Large Language Models. *arXiv preprint arXiv:2304.07619* (2023).
- [14] Igor Makarov and Antoinette Schoar. 2020. Trading and Arbitrage in Cryptocurrency Markets. *Journal of Financial Economics* 135, 2 (2020), 293–319.
- [15] Stefan Nagel. 2021. *Machine Learning in Asset Pricing*. Princeton University Press.
- [16] Dimitris N. Politis and Joseph P. Romano. 1994. The Stationary Bootstrap. *J. Amer. Statist. Assoc.* 89, 428 (1994), 1303–1313.
- [17] Halbert White. 2000. A Reality Check for Data Snooping. *Econometrica* 68, 5 (2000), 1097–1126.
- [18] Hongyang Yang, Xiao-Yang Liu, and Christina Dan Wang. 2023. FinGPT: Open-Source Financial Large Language Models. In *FinLLM Workshop at IJCAI*.
- [19] Shunyu Yao, Jeffrey Zhao, Dian Yu, Nan Du, Izhak Shafran, Karthik Narasimhan, and Yuan Cao. 2023. ReAct: Synergizing Reasoning and Acting in Language Models. In *ICLR*.
- [20] Yangyang Yu, Haohang Li, Zhiyang Chen, Yuechen Jiang, Yang Li, Denghui Zhang, Rong Liu, Jordan W. Suchow, and Khaldoun Khashanah. 2024. FinMem: A Performance-Enhanced LLM Trading Agent with Layered Memory and Character Design. *AAAI Symposium Series* 3, 1 (2024), 595–597.
- [21] Wentao Zhang, Lingxuan Zhao, Hao Xia, Shuo Sun, Jiaye Sun, Molei Qin, Xinyi Li, Yuqing Zhao, Yilei Zhao, Xinyu Cai, et al. 2024. A Multimodal Foundation Agent for Financial Trading: Tool-Augmented, Diversified, and Generalist. In *KDD*.